SeaTable

How to Run Your Own AI

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Running Your Own AI

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It's just a server. With a fat GPU.

Our Stack

Infrastructure

- **Hetzner GPU Server**
- Fixed monthly cost, predictable

Serving

- **vLLM**
- Docker

Model

- **Gemma 3 12B**
- Multimodal: text + images
- 12B parameters

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AI Automations in SeaTable

Trigger

When a row is added

Actions

Run AI

+ Add action

Action

Run AI

Table

Tickets

Function

Classify

Classifies records from one or more columns. You can control the categories, rules, and output column (single or multiple-select) by adjusting the requirements.

Input columns

Ticket

Output column

Category

Classification prompt

Classify this ticket into:
- Billing
- Technical Issue
- Feature Request
Account

The real challenges

Prompting is everything

You want: **"Classify"**

Model returns: *"The category for this ticket is: Billing"*

Model selection trap

- 4B: not reliably
- **12B: the sweet spot**
- 27B: gets to slow

Monitoring is mandatory

Token usage, response times, error rates — without monitoring you're flying blind.

Setting up the hardware is the easy part. The real challenges are prompting, model selection, and monitoring.

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Then We Wanted More

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From automation to conversation

> *"Which projects are overdue? Group them by owner."*

The AI answers — not with a guess, but with real data from your base.

Which opportunities are still active? Group them by owner and show only the two most valuable opportunities for each owner.

Perfect! Now I'll present the active opportunities grouped by owner, showing only the top 2 most valuable opportunities for each:

Active Opportunities by Owner (Top 2 Most Valuable Each)

Steve Rogers

Opportunity	Status	Estimated Value
Product redesign	Proposal	$282,013
Multilanguage concept	Proposal	$123,982

Hulk

Opportunity	Status	Estimated Value
Marketing campaign	Qualification	$34,020
Animation Layout	Proposal	$6,343

Thor

Opportunity	Status	Estimated Value
Seo optimization	Qualification	$88,000
Happiness by design	Proposal	$12,902

Summary: There are 15 active opportunities total (excluding closed deals), with the highest value opportunities belonging to Steve Rogers ($282K and $124K) and Thor having a valuable SEO optimization opportunity at $88K.

Sonnet · **4539 tokens** (3988 in / 551 out) · cached: 9672

3 tool calls ▸

The smart chatbots were ... dump on this...

ChatGPT

Share

Please create a new ticket in my seatable base "Tickets".

I don't have access to any connected resources or tools for your SeaTable account right now, so I'm unable to directly create a ticket in your **"Tickets"** base.

If you'd like, I can:

- Draft the ticket content for you (title, description, priority, tags, etc.) so you can paste it into SeaTable.
- Help you generate a SeaTable API script (Python or curl) that will create the ticket automatically.
- Walk you step-by-step through creating an automation inside SeaTable to do this.

Just tell me what details the new ticket should include (e.g., title, description, priority, assignee), and I'll prepare everything for you.

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MCP Changes Everything

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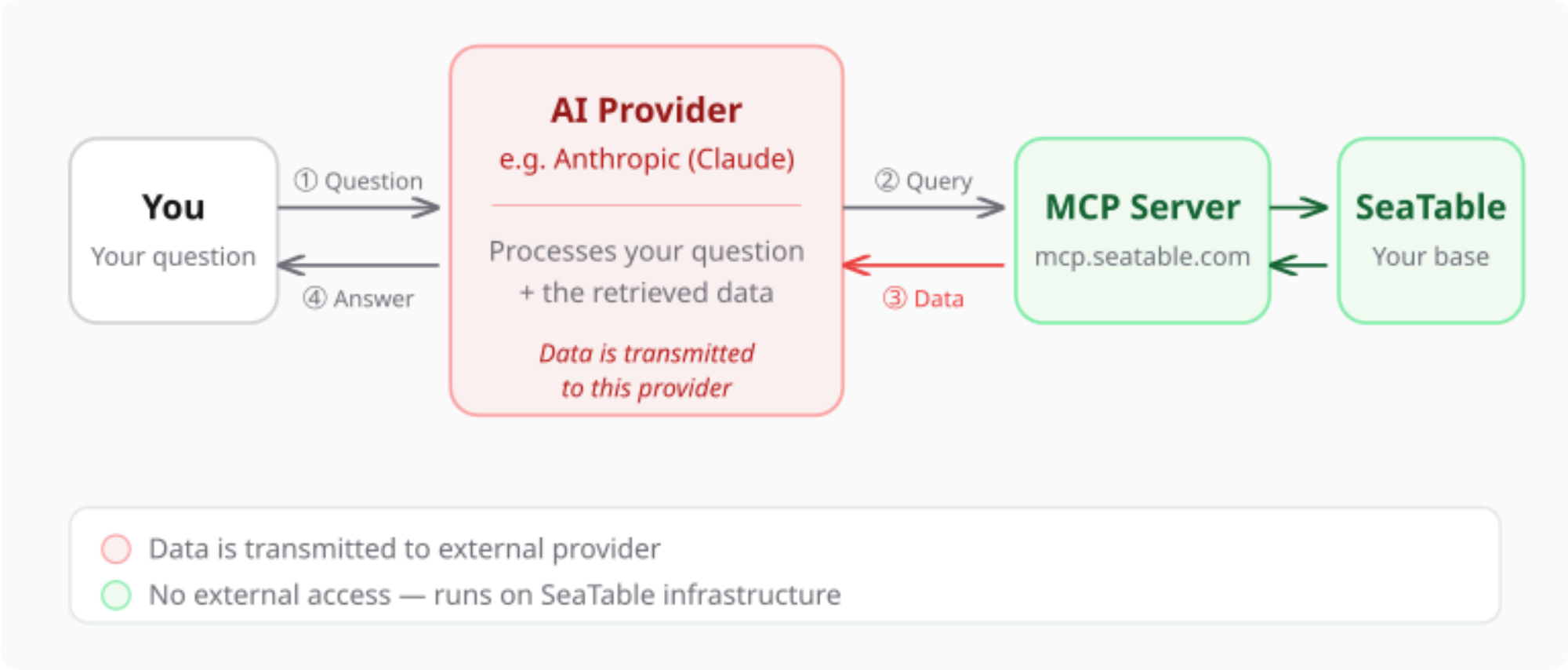
How MCP works



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Without MCP vs. With MCP

	Without MCP	With MCP
Behavior	AI guesses	AI asks
Data	Hallucinations	Real data
Response	*"I don't know your data"*	*"Let me check table X"*
Role	Passive text generator	Active agent

Two-Stage Architecture for Token Optimization

Stage 1 — Tool Selection

- Only category names, no full schemas
- Last 3 messages only
- Max 256 output tokens
- Result: `["read", "sql"]`

Cost: minimal

Stage 2 — Execution

- Only selected tools with full schemas
- Full chat history (max 10 messages)
- Loop up to 10 iterations
- Full answer with data

Cost: only what's needed

Result: ~40–60% fewer tokens per interaction

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The Reality Check

Self-hosting hits its limits

The Capability Problem

MCP requires reliable **tool-calling**:

- Recognize when a tool is needed
- Format the call correctly
- Interpret the result
- Decide if another call is needed

Gemma 3 12B fails at **Stage 1** already — returns prose instead of clean JSON.

The Speed Problem

One chat answer = **3–5 tool calls**
Each call = inference + execute + process

On our GPU: **too slow for interactive use**

The solution: Hybrid Architecture

Automations

- **Process a row**
- Self-hosted **Gemma 3 12B**
- Data stays on our server
- Background processing
- Included in Enterprise plan

AI Chatbot

- **Chat with your data**
- Bring Your Own Model
- Claude (Haiku, Sonnet)
- OpenAI (GPT-4o, o4-mini)
- Mistral (Large, Small)
- User brings API key

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Takeaways

Three things to remember

1. Self-hosting works

For defined AI tasks, running your own LLM is feasible and practical.

2. MCP is the game-changer

It's the difference between AI that guesses and AI that works with your data.

3. The future is hybrid

Self-host where sufficient, frontier models where necessary. Your architecture must support both.

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Interested in this presentation?

- online: https://christophdb.github.io/cs3-2026-how-to-run-your-own-ai/.
- pdf: https://github.com/christophdb/cs3-2026-how-to-run-your-own-ai/blob/main/Slides.pdf